

Quiz 3

Course: Theory of Automata

Marks: 15

Section: BCS-3J

Roll no. _____

Q1: Check if the strings “abbba” and “aabaa” belong to the Context-Free Grammar (CFG) where:

$$S \rightarrow aAB$$

$$A \rightarrow bA \mid b$$

$$B \rightarrow aB \mid a$$

Q2: Derive the string “baabbaab” using the Context-Free Grammar (CFG):

$$S \rightarrow aSbS \mid bSaS \mid \gamma$$

Q3: Design a Context-Free Grammar (CFG) for the language $L = \{a^n b^m \mid n > m\}$ where $n, m > 0$.

Q4: Eliminate the γ -productions (null productions) from the Context-Free Grammar (CFG) where:

$$S \rightarrow ABA$$

$$A \rightarrow aA \mid \gamma$$

$$B \rightarrow bB \mid \gamma$$